



# Quality Assurance Report

| CUSTOMER(고객명) | DESCRIPTION(품명)  | LOT NO(로트번호)      | ISSUED DATE(발행일자) |
|---------------|------------------|-------------------|-------------------|
| 신흥 SEC        | 0.4*30           | 20250502 -01 ~ 06 | 2025-05-02        |
| 규격(재질)        |                  | LOT SIZE(수량)      |                   |
| 0.4*30 AL     | Ni 1.5 $\mu$ m ~ | 342 Kg            |                   |

## I. VISUAL INSPECTION (Sampling Plan : n=100mm/Reel, 5Reel/Shipping or n=50ea/5Reel)

| NO | INSPECTION ITEMS  | Ac/Re | RESULT | NO | INSPECTION        |
|----|-------------------|-------|--------|----|-------------------|
| 1  | DISCOLORATION(변색) | 0/1   | OK     | 6  | NO PLATING(미도금)   |
| 2  | STAIN(일록)         | 0/1   | OK     | 7  | PEELING(도금 벗겨짐)   |
| 3  | RUST(부식)          | 0/1   | OK     | 8  | LUMP(도금물침)        |
| 4  | CONTAMINATION(오염) | 0/1   | OK     | 9  | CAMBER(휨)         |
| 5  | BENT(변형)          | 0/1   | OK     | 10 | PLATING BURR (도금) |

t/z  
0.4x30  
① t/z ④ t/z  
② t/z ⑤ t/z  
③ t/z ⑥ t/z  
총 342Kg

## II. DIMENSION INSPECTION (Sampling Plan : n=100mm/Reel, 5Reel/Shipping or n=50ea/5Reel)

| NO | 항목/구분 | SPEC |       |       | 검사주기     | 우리하이피  |       |
|----|-------|------|-------|-------|----------|--------|-------|
|    |       | 규격   | 하한(-) | 상한(+) |          | 1차     | 2차    |
| 1  | 전폭    | 30.0 | -0.1  |       | Reel/Lot | 29.987 | 0.403 |
| 2  |       |      |       |       |          | 29.967 | 0.401 |
| 3  |       |      |       |       |          | 29.994 | 0.402 |
| 4  |       |      |       |       |          | 29.998 | 0.401 |
| 5  |       |      |       |       |          | 29.979 | 0.402 |
| 6  |       |      |       |       |          | 29.990 | 0.403 |
| 7  |       |      |       |       |          |        |       |
| 8  |       |      |       |       |          |        |       |
| 9  |       |      |       |       |          |        |       |
| 10 |       |      |       |       |          |        |       |

## III. PLATING THICKNESS(도금두께) (Sampling Plan : 1y)

| NO | SPECIFICATION |                  | Measurement Data         | RESULT |
|----|---------------|------------------|--------------------------|--------|
| 1  | X-Ray 측정      | Ni 1.5 $\mu$ m ~ | 첨부 : 도금성적서<br>첨부 : 도금성적서 | OK     |

## IV. FUNCTIONAL TEST (Sampling Plan : n=100mm/EA, 2EA/Shipping, c=0)

| NO | INSPECTION ITEMS          | SPECIFICATION                                                                                                          | TEST SAMPLE | RESULT |
|----|---------------------------|------------------------------------------------------------------------------------------------------------------------|-------------|--------|
| 1  | ADHESION TEST<br>(밀착성 시험) | 1.제품을 스카치 Tape에 완전<br>밀착후 45도 각도로 순간적으로<br>떼어낸다.(들뜸 및 벗겨짐)<br>2.테이프 시험이 안되는 제품은<br>니퍼로 90도로 상하각은후 도금들뜸<br>이나 벗겨짐이 없을것. |             | OK     |

|      |  |       |     |     |      |  |
|------|--|-------|-----|-----|------|--|
| 제품도면 |  | 우리하이피 | 작성  | 승인  | 종합판정 |  |
|      |  |       | 김영란 | 조승현 |      |  |
|      |  | 고객    | 작성  | 승인  | 종합판정 |  |
|      |  |       |     |     |      |  |

고객명: 신흥SEC

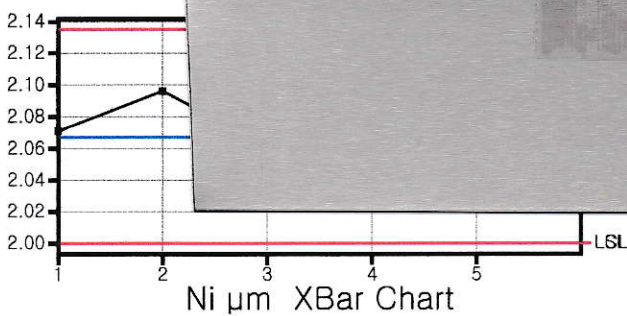
품명 : 0.4X30 초물 LOT NO : 250502-01 DATE : 25.05.02.

| No    | $\mu\text{m}$<br>Ni | Ni tolerance |       |       |
|-------|---------------------|--------------|-------|-------|
|       |                     | 2.032        | 2.067 | 2.102 |
| 1     | 2.071               |              |       |       |
| 2     | 2.096               |              |       |       |
| 3     | 2.062               |              |       |       |
| 4     | 2.085               |              |       |       |
| 5     | 2.057               |              |       |       |
| 6     | 2.032               |              |       |       |
| Mean  | 2.067               |              |       |       |
| S.Dev | 0.022               |              |       |       |

FUNC: MEASURE  
COL : 0.100 mm

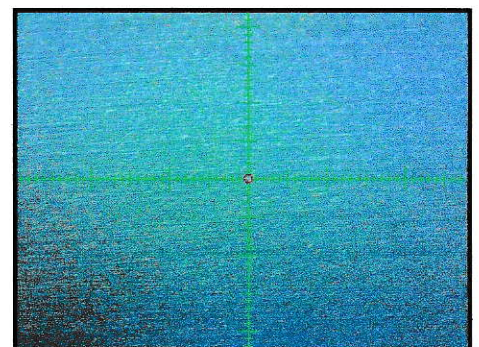
Ni  
—  
Al

SEC : 15 Sec



도금원소 : Ni//Al

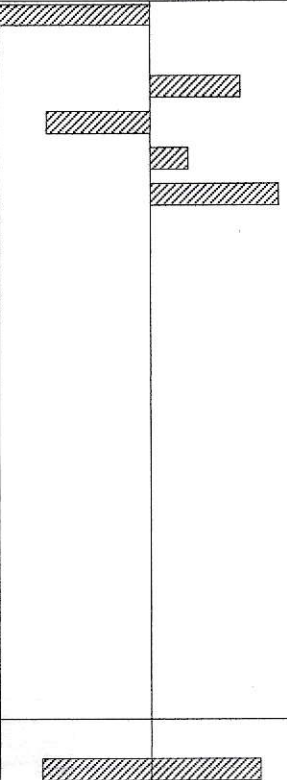
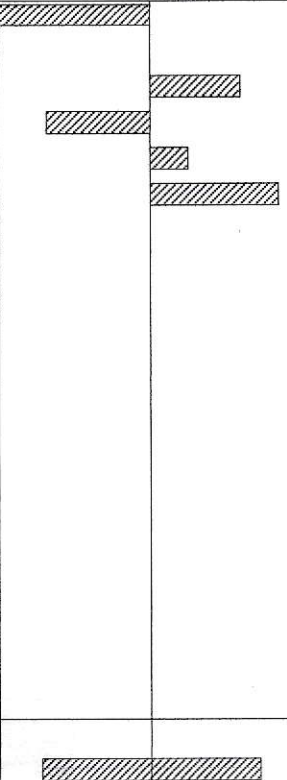
측정값 개수 6  
최대값 2.096  
최소값 2.032  
범위 0.064  
평균값 2.067  
표준편차 0.022  
변동계수 1.088



우리하이피(주)

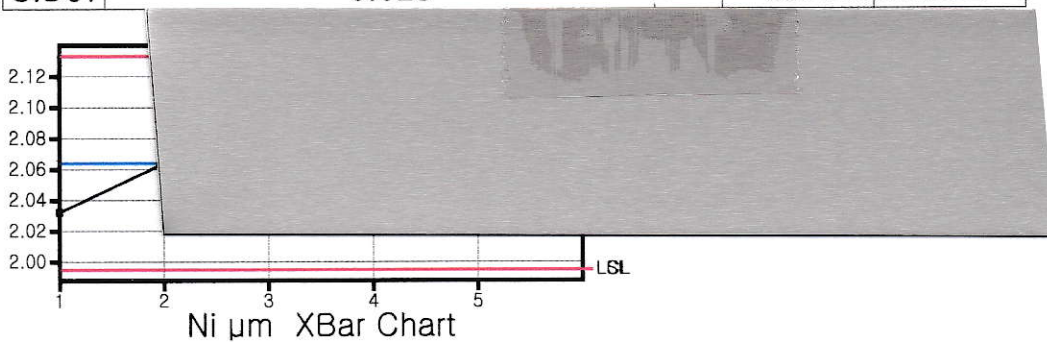
고객명: 신흥SEC

품명 : 0.4X30 초물 LOT NO : 250502-06 DATE : 25.05.02.

| No    | $\mu\text{m}$<br>Ni | Ni tolerance                                                                        |       |       |
|-------|---------------------|-------------------------------------------------------------------------------------|-------|-------|
|       |                     | 2.032                                                                               | 2.064 | 2.096 |
| 1     | 2.032               |  |       |       |
| 2     | 2.064               |                                                                                     |       |       |
| 3     | 2.083               |                                                                                     |       |       |
| 4     | 2.042               |                                                                                     |       |       |
| 5     | 2.072               |                                                                                     |       |       |
| 6     | 2.091               |                                                                                     |       |       |
| Mean  | 2.064               |  |       |       |
| S.Dev | 0.023               |                                                                                     |       |       |

FUNC: MEASURE  
COL : 0.100 mm

$\frac{\text{Ni}}{\text{Al}}$   
SEC : 15 Sec



도금원소 : Ni//Al  
측정값 개수 6  
최대값 2.091  
최소값 2.032  
범위 0.059  
평균값 2.064  
표준편차 0.023  
변동계수 1.118

우리하이피(주)

